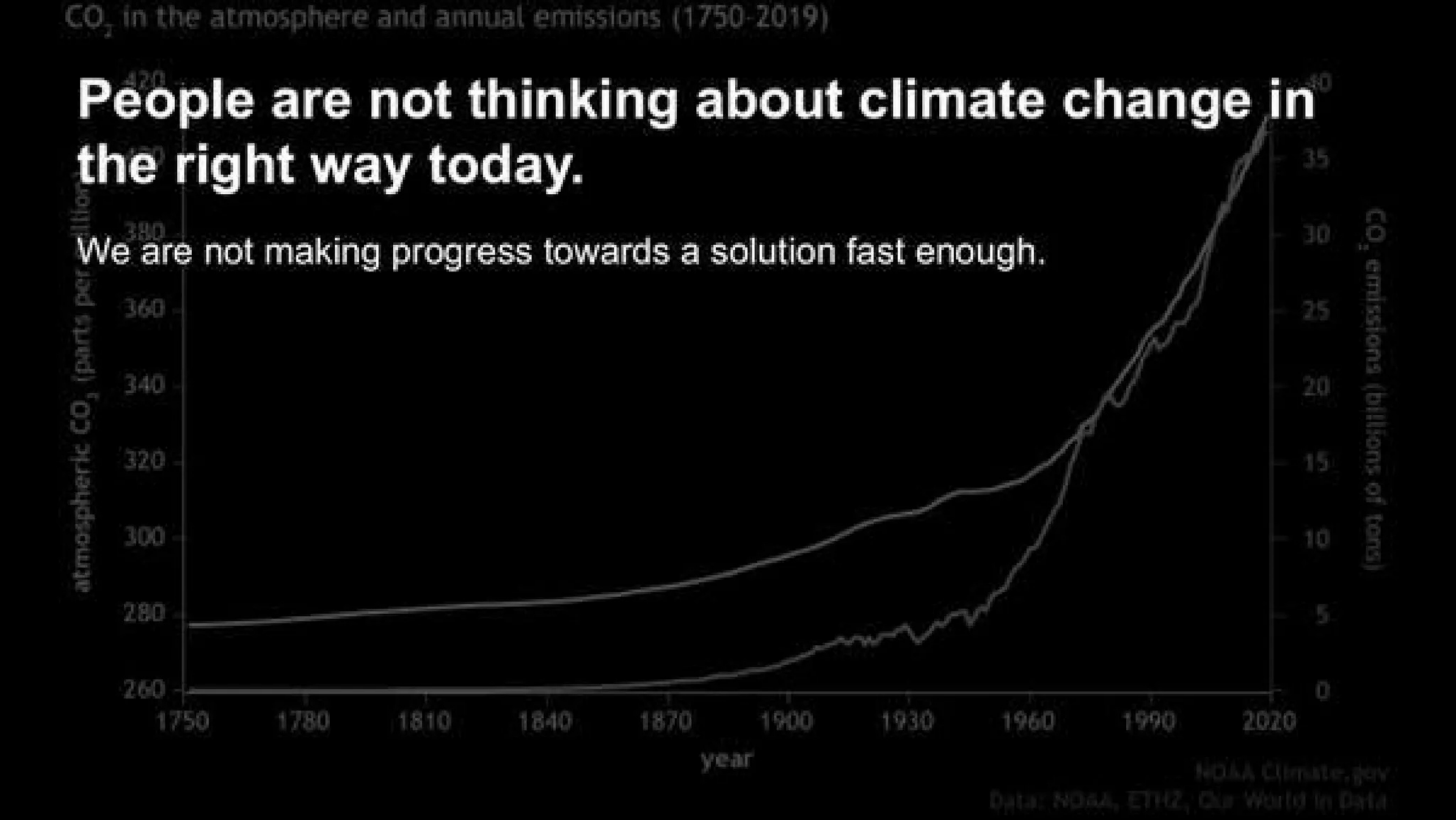

Terraformation
Hyperscaling forest restoration
to solve climate change


CO_2 in the atmosphere and annual emissions (1750-2019)
People are not thinking about climate change in the right way today.
We are not making progress towards a solution fast enough.
atmospheric CO_2 (parts per million)
CO_2 emissions (billions of tons)
420
380
360
340
320
300
280
260
40
35
30
25
20
15
10
5
0
1750
1780
1810
1840
1870
1900
1930
1960
1990
2020
year
NOAA Climate.gov
Data: NOAA, ETHZ, Our World in Data

The key idea everyone is missing is SCALE

The numbers involved are so large that human brains can't comprehend them. As a result, very few people are grasping the full problem or formulating solutions in the context of scalability.

Climate conversations regularly include *very* large numbers, like:

- 45 billion tons of CO_2 emitted per year
- 800 billion tons of extant CO_2 remaining in atmosphere
- 1 trillion trees

Because the problem is incomprehensibly large, most people make the mistake of thinking that what we need is **new technology**. Instead, the solution is to apply **scalability techniques** to proven technology.

What do we believe
that almost everyone else disagrees with?

Solutions employing new technology **will not work** because they will still have to contend with the problems of scalability, plus R&D, plus product risk of new tech,

- or be doomed to comprising less than 0.1% of the solution
- or take too long to scale

Four key lessons

Silicon Valley has learned about solving ultra large-scale problems

#1 Use low-cost, reliable commodity units: Google, Facebook

Ex: Complex mainframes and proprietary new tech doesn't scale, or takes much longer than projected

#2 Going from MVP to global-billions-scale takes **decades**

Ex: Cell phones: first brick phones (1973) to 70% global cell phone ownership (2020)

#3 Scalability has its own set of unique challenges

Ex: SV Eng teams have unique metrics and practices for managing "time to live" -- de-risk everything; avoid new tech, use proven tech

#4 Automate wherever possible, parallelize everything else

Ex: We are decentralizing seed bank infrastructure, massively parallelizing forestry by offering open-source apps and training which turn non-foresters into productive reforestation technicians

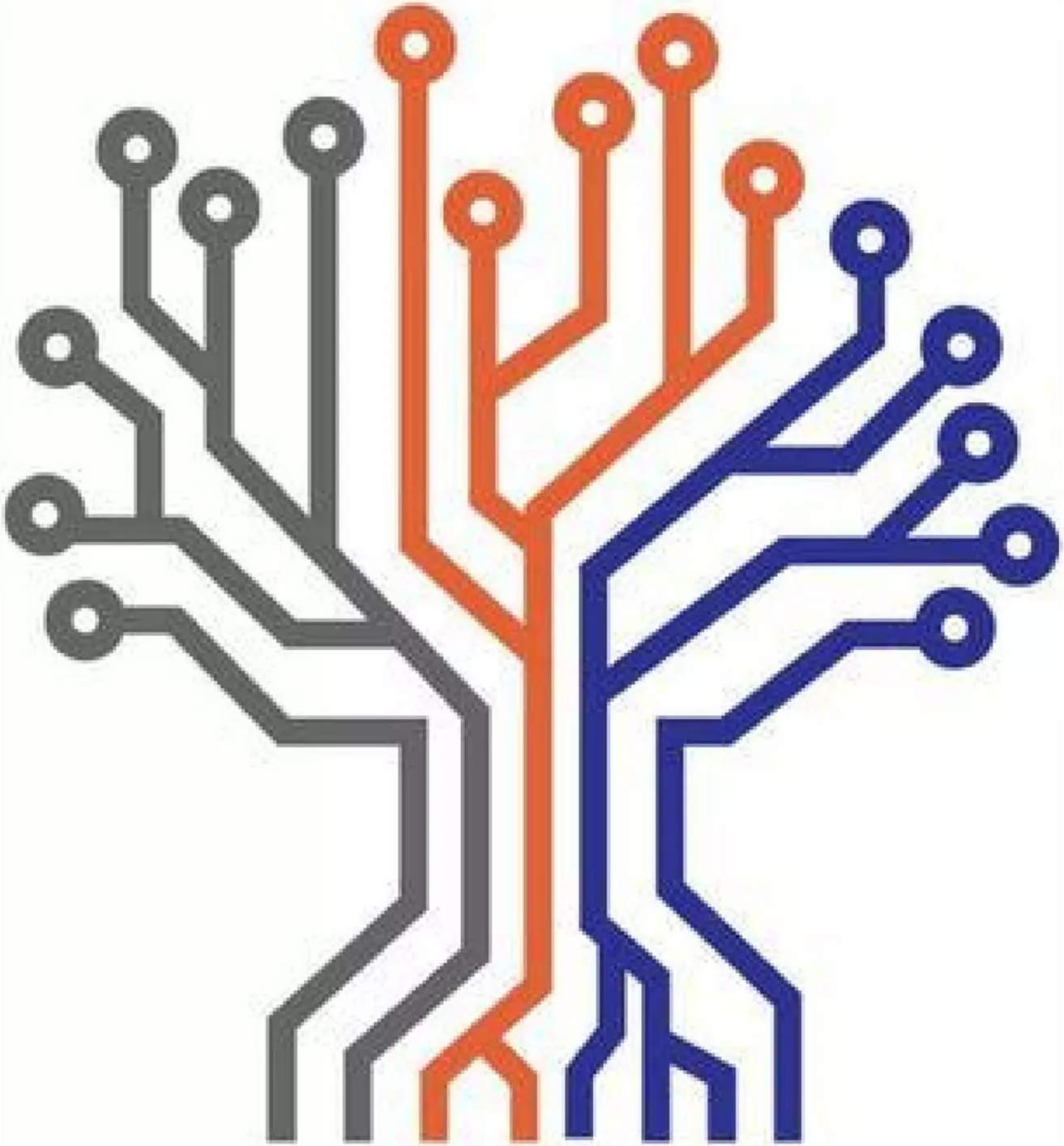


Photo: Terraformation site in North Kohala, HI



Massive reforestation is the lowest-cost, safest, fastest, most effective, and most politically feasible way to remove CO_2 from the atmosphere.

If I told you I invented a new self-replicating carbon sequestration machine that ran on water and sunlight, produced no pollution, could be operated by people around the world, already had "localized" into a thousand customized variants, and some of those variants produced organic food, you'd be asking the terms to our Series A.



Why is this not being pursued?
Because most people don't really understand scalability.



Many people look at trees and conclude, "but there isn't enough land and freshwater to plant enough forests, therefore we cannot use trees to sequester all that carbon."

That is the wrong way to think.



When you have a solution that is superior on multiple dimensions but is held back by X or Y scalability bottleneck, the answer isn't "Ok, let's look for something else entirely," the answer is "Can we solve X and Y bottleneck?" Because then you unlock something extremely valuable.





Solar prices are low enough to make low-emissions desalination economically possible at large scale.

Desalination can tolerate solar intermittency, enabling us to leapfrog the solar transition years ahead of residential/commercial.

We solved the land and freshwater bottlenecks



Scale and Return

This is a solution that can begin scaling immediately. It is not subject to the lab -> product -> scale sequence.

> **Literally:** you give money to a forestry org, and they begin removing CO2 molecules from the atmosphere *that same month*.

This is a solution that's an investment, not a consumption tax.

You're not just paying a sort of tax that goes away: money spent on reforestation eventually pays a return, a huge one in the long term.